Stone–Jordan Exceptional Theory

From the Void by Mirror Climb

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Abstract

We formulate *Stone–Jordan Exceptional Theory* (SJET) from two primitive axioms: the **void** $\mathcal{V}$ (no observable distinction) and a **mirror operator** M that generates structure by involutive doubling. Iterating M on the Stone spectrum of $\mathcal{V}$ climbs the Hurwitz ladder $0 \to \mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to J_3(\mathbb{O})$ and the exceptional cascade $0 \to \mathrm{E}_8 \to \mathrm{E}_8 \times \mathrm{E}_8 \to \mathrm{E}_6 \to \cdots \to 0$; capacity balancing on the mirror-refined graph is the *derived* selection principle, not an independent axiom. At the E_6 rung we prove vacuum selection on EIII, composite $\mathrm{Spin}(10) \times U(1)$ gauge theory with anomaly cancellation, and induced gravitation; postulates P1–P4 complete SM descent. The universe is the stabilized mirror climb $\mathsf{C}^\infty(\mathcal{V})$, equivalently the capacity fixed point on the Albert–Cayley graph.

Keywords: void; mirror operator; Stone duality; exceptional Jordan algebra; E_6 coset; induced gravity.
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Contents

1 Introduction

SJET asks: what is the minimal mathematical seed from which the Standard Model and gravitation can be *derived*, not merely embedded? Our answer is two-fold:

Axiom 1 **The void** $\mathcal{V}$ — no observable distinction, the terminal object of the Stone ladder.

Axiom 2 **The mirror operator** M — an involutive generator that, applied repeatedly, *climbs* the division and exceptional ladders.

Everything else — graphs, Albert algebra, capacity dual, E_6 potential, gauge sector — is **derived** from $\mathsf{M}^n(\mathcal{V})$ and capacity balancing on mirror-refined partitions. Physical postulates P1–P4 (soldering, matter, families, Higgs) remain the honest boundary between proved mathematics and four-dimensional quantum field theory.

1.1 The climb picture

Schematically:

$$\mathcal{V} \xrightarrow{\mathsf{M}} S_{\mathbb{R}} \xrightarrow{\mathsf{M}} S_{\mathbb{C}} \xrightarrow{\mathsf{M}} S_{\mathbb{H}} \xrightarrow{\mathsf{M}} S_{\mathbb{O}} \xrightarrow{\mathsf{M}} J_3(\mathbb{O}) \xrightarrow{\mathsf{M}} \cdots \xrightarrow{\mathsf{M}} \mathcal{V}. \tag{1.1}$$

The bottom and top 0 are the same void: the climb is a *closed* ladder. At each rung, M acts as Boolean complement (discrete), Cayley–Dickson doubling (normed division), or heterotic/Cartan doubling (exceptional Lie). Capacity balancing $\mathcal{C}$ is the *dual mirror*: it selects the unique balanced partition at each level (BAMLP), not a separate axiom.

1.2 Epistemic conventions

- **Theorem** — proved from Axioms 1–2 and standard mathematics.
- **Derived** — explicit construction from mirror climb (no extra postulate).
- **Postulate** — added four-dimensional physics (P1–P4), named once.
- **Conjecture** — motivated but not controlled.

1.3 Conventions

Signature $(-,+,+,+)$; compact E_6 for dynamics; $\Phi \in J_3(\mathbb{O})$ order parameter; $\mathcal{C}$ capacity functional; M mirror operator; $\mathcal{V}$ void. Casimirs: $C_A(\mathrm{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $C_2(\mathbf{16}) = 45/8$. Branchings as in (3.2).

1.4 Organization

Section 2: void and mirror axioms (the only axioms). Section 3: mirror climb of the unified ladder. Section 4: derived graph, Stone lift, and order parameter. Section 5: capacity, potential, vacuum. Sections 6–8: physics under P1–P4. Section 9: ledger. Appendix A: Albert identities.

2 The Void and the Mirror Operator

This section states the *only* primitive axioms of SJET. All graph, Jordan, and Lie data in later sections are derived by mirror climb.

2.1 The void

Definition 2.1 (Void)**.** The *void* $\mathcal{V}$ is the trivial observable algebra: the two-element Boolean algebra $\mathcal{B}_0 = \{\perp, \top\}$ with $\perp < \top$, and its Stone spectrum $S_0 = \mathrm{Spec}(\mathcal{B}_0)$, a single point. The associated division datum is the zero algebra $\{0\}$ of dimension 0. No vertex, no edge, no localisable quantum number exists at $\mathcal{V}$.

Axiom 2.2 (Void)**.** Physical reality is anchored at the void $\mathcal{V}$: the state with no distinguishable observable. All structure arises by departure from $\mathcal{V}$, not by importing a graph or Lie group as primitive data.

2.2 The mirror operator

Definition 2.3 (Mirror operator)**.** A *mirror operator* on a Stone spectrum S is a continuous involution $\mathsf{M} : S \to S$ together with a compatible algebra involution on $C(S)$. At rung n, M_n doubles observable content in one of three modes:

(i) **Discrete (Boolean).** Complement $A \mapsto A^c$ in $\mathcal{B}(\mathcal{G})$; refinement adds a cut, doubling clopen sectors.

(ii) **Arithmetic (Cayley–Dickson).** Normed division doubling $D \mapsto D^{\mathsf{M}}$ with $\dim D^{\mathsf{M}} = 2 \dim D$ for $D \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$; terminal failure at sedenions fixes the $\mathbb{O}$ top.

(iii) **Exceptional (heterotic/Cartan).** Group mirror $G \mapsto G \times G$ (heterotic doubling) or Cartan involution θ on symmetric space G/H (coset mirror $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$).

Axiom 2.4 (Mirror)**.** There exists a functorial mirror M such that the *climb*

$$S_{n+1} \;=\; \mathsf{M}(S_n), \qquad S_0 = S(\mathcal{V}), \tag{2.1}$$

generates the Hurwitz–Albert tower and the exceptional cascade (3.1). The climb is closed: after electroweak breaking the mirror returns to $\mathcal{V}$. Capacity balancing at each rung is the *dual* of M: the unique maximizer $\omega^\star$ of $\mathcal{C}$ on the mirror-refined graph (Theorem 3.8).

Remark 2.5 (Involution law). On a stabilized rung, $\mathsf{M}^2 \cong \mathrm{id}$ up to gauge: Boolean complement squares to the identity; complex/quaternionic conjugation squares to the identity; the coset mirror exchanges $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$ and back.

2.3 The climb operator

Definition 2.6 (Climb operator)**.** The *climb operator* is

$$\mathsf{C} \;=\; \mathrm{Bal}_{\mathcal{C}} \circ \mathsf{M}, \tag{2.2}$$

where $\mathrm{Bal}_{\mathcal{C}}$ assigns the unique capacity maximizer on the mirror-refined partition. The *universe datum* is the profinite fixed point

$$\mathfrak{U}^\star \;=\; \varprojlim_{n\to\infty} \mathsf{C}^n(\mathcal{V}). \tag{2.3}$$

Proposition 2.7 (First rungs from the void)**.** $\mathsf{M}(\mathcal{V})$ *carries one Boolean atom: a single distinction, corresponding to* $\mathbb{R}$ *(order). The next mirror doubles to* $\mathbb{C}$ *(*$\mathrm{U}(1)$ *phase); then* $\mathbb{H}$ *(*$\mathrm{SU}(2)$*),* $\mathbb{O}$ *(*G_2*), and* $H_3(\mathbb{O}) = J_3(\mathbb{O})$ *(*$F_4 \subset \mathrm{E}_6$*).*

Proof. Each arithmetic mirror is Cayley–Dickson doubling; Hurwitz's theorem (Theorem 3.2) proves no further normed division algebra beyond $\mathbb{O}$. The 3×3 Hermitian lift is the unique Jordan packaging of $\mathbb{O}$ at dimension 27. ∎

3 The Mirror Climb: Division and Exceptional Ladders

The ladders are not postulated. They are the orbit of $\mathcal{V}$ under M (Axiom 2.4).

3.1 Unified climb

Definition 3.1 (Unified ladder)**.** The *unified mirror climb* is the sequence

$$\mathcal{V} \xrightarrow{\mathsf{M}} \mathbb{R} \xrightarrow{\mathsf{M}} \mathbb{C} \xrightarrow{\mathsf{M}} \mathbb{H} \xrightarrow{\mathsf{M}} \mathbb{O} \xrightarrow{\mathsf{M}} J_3(\mathbb{O}) \xrightarrow{\mathsf{M}} \mathrm{E}_8 \xrightarrow{\mathsf{M}} \mathrm{E}_8 \times \mathrm{E}_8 \xrightarrow{\mathsf{M}} \mathrm{E}_6 \xrightarrow{\cdots} \mathcal{V}. \tag{3.1}$$

Each arrow is M at the appropriate rung, followed by $\mathrm{Bal}_{\mathcal{C}}$ where a partition is required.

3.2 Hurwitz rung (proved)

Theorem 3.2 (Hurwitz)**.** *Mirror doubling in the normed division category produces only*

$$0,\ \mathbb{R},\ \mathbb{C},\ \mathbb{H},\ \mathbb{O}$$

of dimensions $0, 1, 2, 4, 8$. *The next mirror* $\mathsf{M}(\mathbb{O})$ *fails (sedenions): the climb* must *exit to the Jordan rung* $J_3(\mathbb{O}) = H_3(\mathbb{O})$*, dimension* 27.

Proposition 3.3 (Terminal Jordan rung)**.** $J_3(\mathbb{O})$ *is the unique formally real Jordan algebra compatible with the* $\mathbb{O}$ *mirror at the terminal division step. The Stone limit* $\varprojlim_n \mathsf{M}^n(\mathcal{V})$ *stabilizes at* 27 *coordinates.*

3.3 Exceptional rung

Proposition 3.4 (Exceptional mirror steps)**.** *At the Jordan rung,* M *acts as:*

(i) E_8*: maximal automorphism mirror of the* $\mathbb{O}$ *measure before projection to* $J_3(\mathbb{O})$*;*

(ii) $\mathrm{E}_8 \times \mathrm{E}_8$*: heterotic product mirror (left/right sectors);*

(iii) E_6*: rank-one capacity-balanced cell on* $J_3(\mathbb{O})$*;*

(iv) $\mathrm{E}_6 \to H = \mathrm{Spin}(10) \times U(1)$*: Cartan mirror (coset EIII);*

(v) $\cdots \to \mathcal{V}$*: collapse after electroweak breaking.*

Proposition 3.5 (Branchings at E_6)**.** *Under* $H = \mathrm{Spin}(10) \times U(1)$,

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \tag{3.2}$$

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \tag{3.3}$$

and $\mathcal{M} = \mathrm{E}_6/H = \mathrm{EIII}$ *has* $\dim_{\mathbb{R}} \mathcal{M} = 32$. *The coset mirror is* $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$.

Remark 3.6 (Heterotic parallel). $\mathrm{E}_8 \times \mathrm{E}_8$ matches the heterotic gauge group; $h^1(\hat{Q}, V_{16}) = 3$ [11] corroborates three generations (Remark 8.9). **Conjecture.**

3.4 Capacity as dual mirror

Definition 3.7 (Capacity functional)**.** On a mirror-refined graph $\mathcal{G}$ with costs c_i and workload μ,

$$\mathcal{C}(\omega) = -\sum_{a\in V} \mu(a) \log\Big(\sum_i e^{c_i(a)-\omega_i}\Big). \tag{3.4}$$

This is the *dual mirror* of M: where M doubles sectors, $\mathcal{C}$ balances them.

Theorem 3.8 (Concavity and dual existence)**.** *$\mathcal{C}$ is strictly concave on $\{\omega : \sum_i \omega_i = 0\}$. For targets $m_i > 0$, $\sum_i m_i = 1$, there exists unique $\omega^\star$ with $\partial\mathcal{C}/\partial\omega_i(\omega^\star) = m_i$.*

4 Derived Arena: Graph, Albert Algebra, Order Parameter

This section contains no new axioms. The Albert–Cayley graph, Stone lift, and order parameter are **derived** from $\mathsf{C}^\infty(\mathcal{V})$.

4.1 Albert–Cayley graph

Definition 4.1 (Albert–Cayley graph)**.** At the $J_3(\mathbb{O})$ rung of (3.1), the mirror-refined graph $\mathcal{G}_{\text{Albert}} = (V, E, w)$ has $V = \{1, \ldots, 27\}$ (basis of $J_3(\mathbb{O})$), edges where $\partial_a\partial_b \operatorname{Tr}(X^2) \not\equiv 0$, uniform μ, and territory generators z_1, z_{10}, z_{16} for the sectors of (3.2).

Proposition 4.2 (BAMLP partition)**.** *Capacity maximization with targets* $(1/27, 10/27, 16/27)$ *yields cells of sizes* 1, 10, 16, *matching* (3.2).

4.2 Stone lift

Theorem 4.3 (Profinite Stone limit)**.** *Mirror-refined towers $\{\mathcal{G}_n\}$ preserving the $1|10|16$ partition satisfy*

$$\operatorname{Lift}(\mathcal{G}_\infty) \;\cong\; J_3(\mathbb{O}), \qquad S_\infty = \varprojlim_n \operatorname{Spec}(\mathcal{B}(\mathcal{G}_n)).$$

4.3 Order parameter (derived)

Definition 4.4 (Order parameter)**.** The *order parameter* is the smooth section $\Phi : M^4 \to J_3(\mathbb{O})$ transforming in **27** of compact E_6. It is the continuum limit of coordinates on $\operatorname{Lift}(\mathcal{G}_\infty)$, not an independent axiom.

Theorem 4.5 (Invariants and orbits)**.** *Only $N(X) = \det X$ is E_6-invariant (up to character); $Q(X) = \operatorname{Tr}(X^2)$ is F_4-invariant only. Real orbits are classified by* $\operatorname{rank} X$ *and* $N(X)$; $\operatorname{rank} X \le 1 \iff X^\# = 0$ *(Appendix A.2).*

Proposition 4.6 (Vacuum manifold)**.** *The breaking orbit is $\mathcal{M} = \mathrm{E}_6/H = \mathrm{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$, with coset tangent $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$.*

5 Dynamics: Capacity, Potential, and σ-Model

Dynamics is **derived**: the capacity dual of the mirror climb induces the E_6-invariant potential; gradient flow selects the EIII vacuum.

5.1 Derived potential

Definition 5.1 (Continuum potential)**.** The SJET potential on $J_3(\mathbb{O})$ is

$$V(X) = \alpha \langle X^{\#}, X^{\#}\rangle + \beta\big[\mathrm{Tr}(X^2) - c_2\big]^2 + \gamma\ \mathrm{Tr}(X^2)^k, \qquad \alpha, \beta, \gamma > 0,\ k \geq 2 \text{ even}, \tag{5.1}$$

the profinite limit of the graph penalty under $\mathrm{Bal}_{\mathcal{C}}$ on $\mathcal{G}_{\mathrm{Albert}}$.

Theorem 5.2 (Potential identification)**.** *V is the unique E_6-invariant potential with quartic $\langle X^{\#}, X^{\#}\rangle$ vanishing on rank ≤ 1, induced by the mirror-dual capacity on $\mathcal{G}_{\mathrm{Albert}}$.*

5.2 Vacuum selection

Lemma 5.3 (Sharp locus)**.** *$\langle X^{\#}, X^{\#}\rangle \geq 0$ with equality iff $\mathrm{rank}\, X \leq 1$.*

Theorem 5.4 (Global minimum on EIII)**.** *V is coercive on compact E_6. Global minima lie on the rank-one orbit $\mathcal{M} = \mathrm{EIII}$; the origin is a strict local maximum.*

Corollary 5.5 (Tree-level Goldstones)**.** *$\mathrm{Hess}\, V(\Phi_\star) = 0$ on $\mathfrak{m}$: 32 exact Goldstones at tree level.*

5.3 σ-model action

$$S[\Phi] = \int d^4x\, \sqrt{|g|}\, \left[\frac{1}{2} K_{AB}\, \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi)\right], \tag{5.2}$$

with K the unique H-invariant metric on $\mathfrak{m}$ (Theorem 6.2).

5.4 Universe as mirror fixed point

Theorem 5.6 (Universe as $\mathsf{C}^\infty(\mathcal{V})$)**.** *Under Axioms 2.2–2.4 and Postulates P1–P4, the universe datum $\mathfrak{U}^\star$ of (2.3) satisfies:*

(i) $\omega^\star$ maximizes $\mathcal{C}$ on $\mathcal{G}_{\mathrm{Albert}}$;

(ii) $\Phi_\star \in \mathcal{M}$ minimizes V;

(iii) $g_{\mu\nu}$ is the soldered Minkowski metric (P1);

(iv) $\mathcal{H}$ is the three-family SM (P2–P4).

Proof. (i)–(ii) from Theorems 3.8 and 5.4. (iii)–(iv) from postulates. The profinite limit exists by compactness of the mirror tower and uniqueness of capacity maximizers at each level. □

6 Emergent Spacetime and Time

Four-dimensional Lorentzian geometry is not a consequence of the E_6 coset metric alone. Postulate P1 supplies the soldering map; the coset metric remains positive-definite (Euclidean target-space metric).

6.1 Coset metric and irreducibility

Theorem 6.1 (Real-irreducibility of $\mathfrak{m}$)**.** *As a real H-module, $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is irreducible of* complex type*:* $\mathrm{End}_H(\mathfrak{m}) \cong \mathbb{C}$*.*

Proof. Complexifying, $\mathfrak{m}_{\mathbb{C}} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ with non-isomorphic summands exchanged by conjugation. A real submodule would be a complex submodule invariant under conjugation; Schur's lemma on each summand forbids nontrivial proper submodules. ☐

Theorem 6.2 (Unique coset metric)**.** *The restriction $K|_{\mathfrak{m}}$ of the Killing form is the unique H-invariant positive-definite inner product on $\mathfrak{m}$.*

Proof. H-invariant bilinear forms correspond to H-invariant maps $\mathfrak{m} \otimes \mathfrak{m} \to \mathbb{R}$. By Schur, such maps factor through the trivial representation; complex type with opposite $U(1)$ charges forbids symmetric invariants $\mathbf{16}_{-3} \otimes \mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3} \otimes \overline{\mathbf{16}}_{+3}$, leaving only the cross pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3} \supset \mathbf{1}_0$, unique up to scale. ☐

Remark 6.3. $K|_{\mathfrak{m}}$ is **positive-definite**, not Lorentzian. Signature $(-,+,+,+)$ is carried by the soldering form, not by the Killing restriction.

6.2 Postulate P1: soldering

The 2×2 Hermitian-octonionic block embeds as $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ with det the Minkowski norm; $\mathrm{SL}(2, \mathbb{O}) \cong \mathrm{Spin}(1, 9)$.

Postulate 6.4 (Soldering and dimension, P1)**.** There exists a soldering map σ identifying four clock-field directions e^a_μ $(a = 0, 1, 2, 3)$ with a four-dimensional subspace of $H_2(\mathbb{O})$ carrying $\eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$. The world manifold has $d = 4$. The vierbein emerges from the Maurer–Cartan $\mathfrak{m}$-component (Section 7).

Proposition 6.5 (Emergent vierbein)**.** *For a coset section $g(x) \in \mathrm{E}_6$ with $g^{-1}\partial_\mu g = A_\mu + e_\mu$, the $\mathfrak{m}$-component defines a composite vierbein. Under P1, $\eta_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}$ has signature $(-,+,+,+)$.*

6.3 Emergent time

Proposition 6.6 (Page–Wootters clock)**.** *In the semiclassical regime of induced gravity (Theorem 7.4), the Hamiltonian constraint $H_{\mathrm{phys}}|\psi\rangle = 0$ holds on physical states. The soldered timelike clock Φ^0 defines internal time; conditional evolution $U(\tau) = e^{-iH_\tau \tau}$ on physical states recovers standard quantum dynamics.*

Remark 6.7 (OS reconstruction). Euclidean correlators follow from (5.2). Lorentzian unitary dynamics requires Osterwalder–Schrader reconstruction along Φ^0. **Conjecture:** the σ-model satisfies reflection positivity.

7 Gauge Sector and Induced Gravitation

7.1 Composite connection

The Lie algebra splits H-equivariantly as $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ and $\mathfrak{m}$ as in (3.3).

Theorem 7.1 (Composite gauge connection)**.** *For $g(x) \in \mathrm{E}_6$, write $g^{-1}\partial_\mu g = \mathcal{A}_\mu + e_\mu$ with $\mathcal{A}_\mu \in \mathfrak{h}$, $e_\mu \in \mathfrak{m}$. Under $g \mapsto gh^{-1}$, $h \in H$,*

$$\mathcal{A}_\mu \mapsto h\mathcal{A}_\mu h^{-1} - (\partial_\mu h)h^{-1},$$

and the curvature $\mathcal{F}_{\mu\nu} = \partial_{[\mu}\mathcal{A}_{\nu]} + [\mathcal{A}_\mu, \mathcal{A}_\nu]$ transforms as an H field strength. All 46 generators of $\mathfrak{h}$ remain massless because H is unbroken.

Proof. Standard transformation of the canonical connection on the reductive homogeneous bundle $\mathrm{E}_6 \to \mathcal{M}$. □

Proposition 7.2 (Induced Yang–Mills)**.** *At one loop, integrating out massive coset modes generates $S_{\mathrm{YM}} = \frac{1}{4\hat{g}^2}\int \mathrm{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})$ with $\hat{g}^{-2} \propto f^2\mu^2/(16\pi^2)$, $[f] = \mathrm{mass}$.*

7.2 Anomaly cancellation

Anomalies are computed with Dynkin indices $A_{\mathbf{16}} = A_{\mathbf{10}} = 1$, $A_{\mathbf{1}} = 0$ and $U(1)$ charges from (3.2).

Theorem 7.3 (Green–Schwarz cancellation)**.** *The cubic anomaly $\mathcal{A}_{U(1)^3}$ vanishes: $16 \cdot 1^3 + 10 \cdot (-2)^3 + 1 \cdot 4^3 = 0$. The mixed anomaly $\mathcal{A}_{U(1)\cdot \mathrm{Spin}(10)^2}$ has coefficient -1: $1 \cdot 1 + (-2) \cdot 1 + 4 \cdot 0 = -1$, and is cancelled by a Green–Schwarz term $\propto \mathrm{tr}(F \wedge F)\, B$ with axion B from the $\mathbf{1}_{+4}$ sector.*

Proof. Direct arithmetic on charges and indices; see [6] for the mechanism. □

7.3 Induced gravity

Theorem 7.4 (Sakharov induced gravity)**.** *In a heat-kernel scheme, only massive modes contribute to the a_2 coefficient. With 28 physical massive coset scalars ($s = +1$, $m^2 = \mu^2$ canonically),*

$$M_{\mathrm{Pl}}^2 = \frac{1}{16\pi^2} \cdot \frac{1}{6} \sum_{\text{massive}} s_i m_i^2 \log\frac{\Lambda^2}{m_i^2} \approx \frac{28}{6} \cdot \frac{\mu^2}{16\pi^2} \log\frac{\Lambda^2}{\mu^2} > 0. \tag{7.1}$$

The four soldering directions are protected by diffeomorphism and local Lorentz Ward identities, not by $\mathrm{Hess}\, V$*.*

Remark 7.5 (Cosmological constant). The a_0 term yields an uncancelled Λ^4 vacuum energy. SJET does not solve the cosmological constant problem; Λ_{cc} is a tuned relevant coupling.

7.4 Renormalization group

Theorem 7.6 (Gauge β-function)**.** *With canonical dimensionless coupling $\hat{g}^2$,*

$$\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2}\hat{g}^4, \qquad b_0 = \tfrac{11}{3}C_A - \tfrac{4}{3}T\,n_F - \tfrac{1}{6}T\,n_S. \tag{7.2}$$

For three chiral $\mathbf{16}$*s ($n_F = 3$) and $n_S = 28$ coset scalars:*

$$b_0(3\,\mathrm{gen}) = \tfrac{11}{3}(8) - \tfrac{4}{3}(2)(3) - \tfrac{1}{6}(2)(28) = \frac{88 - 24 - 28}{3} = 12. \tag{7.3}$$

The one-generation anchor ($n_F = 1$, $n_S = 2$) gives $b_0(1\,\mathrm{gen}) = 26$. The gauge sector is asymptotically free ($\hat{g}_\star^2 = 0$).

Conjecture 7.7 (Gravitational UV fixed point)**.** *A truncated RG gives $\beta_\kappa = 2\kappa - (5/48\pi^2)\kappa^2$ with UV-attractive $\kappa_\star = 96\pi^2/5$. This one-loop result is indicative, not rigorous [9].*

8 Standard Model Descent

Matter and electroweak symmetry breaking are **postulated** (P2–P4). Anomaly cancellation is a **theorem** once the **27** content is fixed.

8.1 Postulates P2–P4

Postulate 8.1 (Chiral matter, P2)**.** Each family furnishes one chiral **16** of Spin(10) with charges from (3.2). No mirror $\overline{\mathbf{16}}$ partners.

Postulate 8.2 (Three families, P3)**.** Horizontal $SU(3)_F$ acts on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$; the three factors are three chiral **16**s. Three generations are *not* derived from Axioms I–III.

Postulate 8.3 (GUT–Higgs and Yukawa, P4)**.** Breaking proceeds via $\mathbf{45}_H$ or $\mathbf{54}_H$ (Spin(10) $\to$ $SU(5) \times U(1)_X$), $\mathbf{126}_H$ (breaks $B - L$), and $\mathbf{10}_H$ (electroweak). Yukawas $y_{ij}\,\mathbf{16}_i\mathbf{16}_j H$ generate fermion masses; type-I seesaw from $\mathbf{126}_H$ gives $m_\nu \sim m_D^2/M_R$.

Postulate 8.4 (Flavon hierarchy, P3′)**.** Sequential $SU(3)_F$ breaking by flavon VEVs generates Yukawa hierarchy and CKM/PMNS mixing.

8.2 Descent chain

Theorem 8.5 (SM gauge structure under P4)**.** *Assume P4. The breaking chain*

$$\mathrm{E}_6 \xrightarrow{V} \mathrm{Spin}(10) \times U(1) \xrightarrow{\mathbf{45}_H} \mathrm{SU}(5) \times U(1)_X \xrightarrow{\mathbf{126}_H} \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y \xrightarrow{\mathbf{10}_H} \mathrm{SU}(3)_c \times U(1)_{\mathrm{em}}$$

yields the SM gauge group with GUT normalization $\alpha_1 = (5/3)\alpha_Y$.

Per family $\mathbf{16} = (q, u^c, d^c, \ell, e^c, \nu^c)$; $\mathbf{10}_H$ supplies $v = 246\,\mathrm{GeV}$.

Remark 8.6 (Yukawa source term). A bare coupling $\bar{\psi}\,\Phi^a T_a\,\psi$ with Φ in $\{\mathbf{16}, \overline{\mathbf{16}}\}$ is *not* H-invariant and does not generate masses at the $\Phi_\star$ vacuum. Masses require P4 Higgs multiplets [7].

8.3 Scalar spectrum

Theorem 8.7 (Tree-level Goldstones)**.** *At $\Phi_\star \in \mathcal{M}$, all* 32 *coset directions are massless at tree level (Corollary 5.5).*

Theorem 8.8 (Radiative degeneracy)**.** *The Coleman–Weinberg potential [10] is H-invariant. Since $\mathfrak{m}$ is a single real-irreducible H-module (Theorem 6.1), Schur gives one eigenvalue on all* 28 *physical massive scalars; canonical mass $m^2 = \mu^2$.*

Remark 8.9 (Heterotic corroboration). The heterotic $\mathbb{Z}_5$ quintic gives $h^1(\hat{Q}, V_{16}) = 3$ [11]. **Conjecture**, not a derivation of P3.

9 Verification Ledger

9.1 Status table

9.2 Algebraic checks

Anomalies: $16 - 80 + 64 = 0$; mixed $1 - 2 + 0 = -1$. β: $b_0 = 12$. Dimensions: $78 = 45 + 1 + 16 + 16$; $27 = 1 + 10 + 16$; $32 = 28 + 4$.

Claim	Status	Check	Ref.
Void $\mathcal{V}$ primitive	Axiom	$\mathcal{B}_0 = \{\perp, \top\}$	2.2
Mirror M primitive	Axiom	involutive climb	2.4
Hurwitz ladder	Theorem	dims $0, 1, 2, 4, 8$	3.2
$\mathsf{C}^\infty(\mathcal{V}) \cong J_3(\mathbb{O})$	Theorem	27 stable	4.3
$1\|10\|16$ partition	Prop.	capacity	4.2
Potential from $\mathcal{C}$	Theorem	limit	5.2
EIII vacuum	Theorem	coercivity	5.4
Coset mirror $16 \leftrightarrow \overline{16}$	Theorem	$U(1)$ charge	3.5
Anomaly cancellation	Theorem	$16 - 80 + 64$; GS -1	7.3
$b_0(3) = 12$	Theorem	(7.3)	7.6
$\mathfrak{U}^\star = \mathsf{C}^\infty(\mathcal{V})$	Theorem	under P1–P4	5.6
Soldering, $d = 4$	Postulate	P1	6.4
Matter, 3 families	Postulate	P2–P3	§8.1
GUT–Higgs	Postulate	P4	8.3
$\kappa_\star$	Conjecture	1-loop	7.7
Heterotic $h^1 = 3$	Conjecture	string	8.9

Table 1: Epistemic ledger. Only two axioms: void and mirror.

9.3 Falsifiable predictions

F1. Proton decay bound on M_{GUT}.

F2. Seesaw neutrino scale $m_\nu \sim 0.05\,\mathrm{eV}$.

F3. No fourth generation (mirror climb stabilizes at three $SU(3)_F$ copies).

F4. Common CW mass μ on 28 coset scalars.

F5. Fourth generation would break $b_0 > 0$ window.

10 Conclusions

SJET reduces its primitive content to **two axioms**:

Axiom 1 the **void** $\mathcal{V}$ — no observable distinction;

Axiom 2 the **mirror operator** M — involutive doubling that climbs the division and exceptional ladders.

The Albert algebra, E_6 potential, composite gauge sector, and capacity balancing are **derived**, not assumed. The universe is

$$\mathfrak{U}^\star \;=\; \mathsf{C}^\infty(\mathcal{V}) \;=\; \varprojlim_n \big(\mathrm{Bal}_{\mathcal{C}} \circ \mathsf{M}\big)^n(\mathcal{V}), \tag{10.1}$$

stabilizing at $J_3(\mathbb{O})$ and E_6, with physical contact via P1–P4.

10.1 Open problems

O1. Rigorous functorial definition of M across all rungs in one category.

O2. Derive P3 (three families) from mirror stabilisation alone.

O3. Prove $\mathsf{M}(\mathrm{E}_8) = \mathrm{E}_8 \times \mathrm{E}_8$ from Jordan data without heterotic input.

O4. Gravitational UV fixed point beyond one-loop truncation.

O5. Cosmological constant.

A Albert Algebra Identities

A.1 Matrix model

Elements of $J_3(\mathbb{O})$ are

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \qquad \xi_i \in \mathbb{R},\ x_i \in \mathbb{O}, \tag{A.1}$$

with Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$. The trace inner product is $\langle X, Y \rangle = \mathrm{Tr}(X \circ Y)$ and $Q(X) = \mathrm{Tr}(X^2)$.

A.2 Sharp map and rank

The cubic norm is $N(X) = \det X$ (octonionic determinant). The sharp map is

$$X^\# = \mathrm{adj}(X), \qquad X^\# \circ X = N(X)\, E, \qquad (X^\#)^\# = N(X)\, X, \tag{A.2}$$

with $E = \mathrm{diag}(1,1,1)$. In diagonal form $X = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$, $X^\# = \mathrm{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$. Thus $\mathrm{rank}\, X \leq 1 \iff X^\# = 0$.

A.3 Freudenthal identity

For the quadratic adjoint $X \mapsto X^\#$,

$$(X \circ Y)^\# = (N(X)Y + N(Y)X - X^\# \circ Y^\#), \tag{A.3}$$

which characterizes $J_3(\mathbb{O})$ among cubic vector spaces. Any E_6-invariant quartic vanishing on $\{X^\# = 0\}$ is proportional to $\langle X^\#, X^\# \rangle$.

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